

QUARK MASSES AND FLAVOR MIXING

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The possibility of calculating the flavor mixing angles in the six-quark theory based on the gauge group $SU(2)_L \times SU(2)_R \times U(1)$ is discussed. The t quark mass can be computed in terms of the u , d , s , c , and b quark masses. We estimate $m_t \approx 10 \dots 11$ GeV. Phenomenological implications with respect to the decays of b and t flavored particles are described. The B mesons decay into charmed particles.

1. Introduction

At present all experimental data in weak interaction physics are in agreement with the minimal $SU(2) \times U(1)$ theory [1]. In this framework the leptons and quarks come in left-handed doublets of the weak isospin. At least three such doublets are required by the present experimental findings. There is direct evidence for the existence of three lepton doublets. Two quark doublets are established and the charge $-\frac{1}{3}$ component of the third doublet, the b -quark, has been found. No direct evidence exists thus far for the existence of the t quark as the charge $\frac{2}{3}$ component of the third doublet. Various experimental constraints require the effective t -quark mass to be larger than about eight GeV [2].

In this paper we shall assume that the minimal $SU(2) \times U(1)$ theory describes correctly the observed weak and electromagnetic interactions. Of course, this does not exclude the possibility that the minimal flavor gauge group $SU(2) \times U(1)$ is a subgroup of a larger flavor group, which becomes relevant at energies large compared to the energy corresponding to the W and Z boson masses (W, Z : massive gauge bosons of the minimal $SU(2) \times U(1)$ framework). In fact, we shall assume that $SU(2) \times U(1)$ is a subgroup of the larger gauge group $SU(2) \times SU(2) \times U(1)$. Furthermore we assume that all quarks and leptons are in left-handed doublets of the weak isospin. We shall concentrate mainly on the case of three doublets:

$$\begin{pmatrix} \nu_e & \nu_\mu & \nu_\tau \\ e^- & \mu^- & \tau^- \end{pmatrix}, \quad \begin{pmatrix} u & c & t \\ d & s & b \end{pmatrix}.$$

The weak interactions of the quarks are described by the weak interaction mixing angles; in case of three doublets one is dealing with three Cabibbo-like mixing angles [3] and one phase parameter. The latter leads to a violation of CP invariance. In principle one can also introduce weak mixing angles in the leptonic sector. However, there they have no physical meaning (they can be transformed away) as long as the neutrinos are massless. If the neutrinos have some small masses, the effects due to the weak interaction mixing are confined to rather exotic phenomena (e.g., neutrino oscillations), and we shall not discuss the flavor mixing in the lepton sector any further.

The weak mixing angles arise since in reality the quark states for which the weak interaction is diagonal are not eigenstates of the quark mass matrix. If all quarks of a given charge were degenerate in mass (e.g., zero) the weak mixing angles (e.g., the Cabibbo angle) could be transformed away, i.e., in this case they would have no physical meaning. Thus the weak mixing angles can be regarded as elements of the quark mass matrix. Any attempt to calculate the weak mixing angles must be at the same time an attempt to say something about the quark masses. Recently many authors have studied possible relations between the quark masses and the weak interaction mixing angles [4–14]. In this paper we shall discuss one possible line of argument, namely the one followed in ref. [9], in more detail. We conclude that the three weak mixing angles in the six-flavor case can to a certain extent be calculated in terms of the quark masses. There is no way to determine the CP violating phase. A relation for the t -quark mass is obtained, which gives $m_t \approx 10 \dots 11$ GeV.

Thus far, nobody has proposed a scheme or a theory in which the various quark and lepton masses can be predicted. In the gauge theories of flavor dynamics (e.g., the minimal $SU(2) \times U(1)$ theory) the quark and lepton masses are essentially free parameters of the theory. On the other hand a short look at the observed mass spectrum of the leptons and quarks is enough in order to make clear that there exists a definite order. The fermions seem to come in lepton-quark generations. At present three generations are required for phenomenological reasons:

$$\begin{aligned}
 \text{I} \quad & \begin{pmatrix} \nu_e & | & u[4.2] \\ e^- [0.5] & | & d[7.5] \end{pmatrix}, \\
 \text{II} \quad & \begin{pmatrix} \nu_\mu & | & c[1200] \\ \mu^- [106] & | & s[150] \end{pmatrix}, \\
 \text{III} \quad & \begin{pmatrix} \nu_\tau & | & t[>8000] \\ \tau^- [1780] & | & b[4400] \end{pmatrix}.
 \end{aligned} \tag{1.1}$$

The masses of the fermions are given in the square brackets (in MeV). No masses or limits on the masses for the neutrinos are given, since at present all observed phenomena are consistent with $m(\nu_e) = m(\nu_\mu) = m(\nu_\tau) = 0$. For the u , d , and s quarks we have denoted the mass values obtained using PCAC (see, e.g., ref. [15]).

We should like to add the following remarks.

(a) The classification of the fermions into generations is made on the basis of the masses. All charged fermions which are lighter than 100 MeV belong to generation I. All charged fermions with masses larger than 100 MeV and smaller than 1500 MeV belong to generation II; the ones which are heavier than 1500 MeV belong to generation III. There is no overlap: the fermions of a given generation are heavier than the heaviest fermion of the previous generation and lighter than the ones of the next generation (with the exception of the neutrinos).

(b) The main weak transitions occur between the fermions of a given generation. All other weak transition amplitudes are multiplied by small angles, and in some sense they might be considered as a perturbation.

(c) It seems that the ratio of the masses of two fermions which follow each other in the sequence of the generations are roughly of the same order of magnitude:

$$m_d/m_s \sim m_b/m_t, \quad m_\mu/m_e \sim m_\tau/m_\nu. \quad (1.2)$$

This implies that a logarithmic plot of the fermion masses shows almost straight lines (see fig. 1 and ref. [16]).

In the six-flavor case we can write the flavor mixing in terms of three Euler angles and one phase parameter [3]. We choose the parametrization used in ref. [17]. The weak currents are given by

$$\bar{U}\gamma_\mu(1 + \gamma_5)MD,$$

where $U = (u, c, t)$ and $D = (d, s, b)$, and M stands for the 3×3 unitary matrix

$$\begin{aligned} M &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_2 & s_2 \\ 0 & -s_2 & c_2 \end{bmatrix} \begin{bmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{i\delta} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_3 & s_3 \\ 0 & -s_3 & c_3 \end{bmatrix} \\ &= \begin{bmatrix} c_1 & s_1 c_3 & s_1 s_3 \\ -s_1 c_2 & c_1 c_2 c_3 - s_2 s_3 e^{i\delta} & c_1 c_2 s_3 + s_2 c_3 e^{i\delta} \\ s_1 s_2 & -c_1 s_2 c_3 - c_2 s_3 e^{i\delta} & -c_1 s_2 s_3 + c_2 c_3 e^{i\delta} \end{bmatrix}. \end{aligned} \quad (1.3)$$

Here c_1 stands for $\cos \theta_1$, s_1 for $\sin \theta_1$, etc. The universality constraint, based on the universality of the $(\bar{u}d)$ and $(\bar{u}s)$ currents, implies $\theta_3 \lesssim 15^\circ$. Furthermore there exists a limit for θ_2 based on the estimate of the $K_1 - K_2$ mass difference [17]

$$\theta_2 \lesssim \sqrt{\frac{m_c}{m_t}}.$$

The first angle θ_1 is essentially the Cabibbo angle: $\theta_1 = \theta_C = 13.2^\circ$. The phase parameter δ induces CP violation. It is related to the ϵ parameter by

$$|\epsilon| \approx 2 \cdot 10^{-3} \approx 3s_2 c_2 s_3 \sin \delta.$$

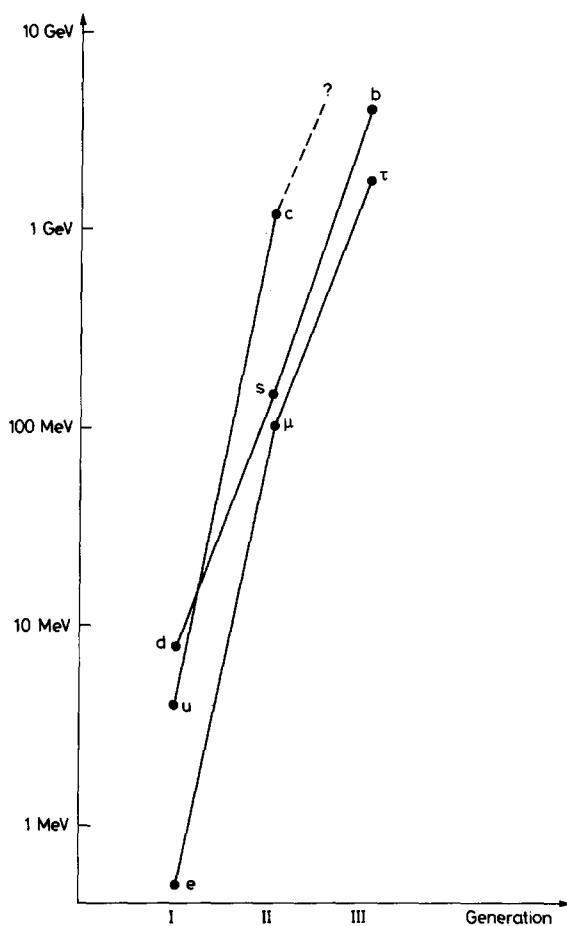


Fig. 1. The quark and lepton masses.

2. The quark mass matrix

If one looks at the spectrum of the quark masses, it is apparent that the c and s quark masses are small compared to the t and b quark masses respectively, and that the u and d quark masses are small compared to the c and s quark masses. Furthermore the weak mixing angles are small. Thus one might expect that the quark mass ratios like m_d/m_s , m_u/m_c , m_c/m_t , m_s/m_b are in some way related to the weak mixing angles. Furthermore there seems to exist a hierarchy of quark masses. One might regard the b and t quark masses as the “driving terms” in the quark mass matrix, and the c and s quark masses as first-order perturbations, and the u and d quark masses as second-order perturbations.

In this paper we shall study a particular ansatz for the quark mass matrix, which was introduced in ref. [9]. We shall assume that the quark mass matrix can be written as follows:

$$\overline{(u_0, c_0, t_0)}_L \begin{pmatrix} 0 & A & 0 \\ A' & 0 & B \\ 0 & B' & C \end{pmatrix} \begin{pmatrix} u_0 \\ c_0 \\ t_0 \end{pmatrix}_R + \text{h.c.},$$

$$|A| = |A'|, \quad |B| = |B'|, \quad (2.1)$$

$$\overline{(d_0, s_0, b_0)}_L \begin{pmatrix} 0 & \tilde{A} & 0 \\ \tilde{A}' & 0 & \tilde{B} \\ 0 & \tilde{B}' & \tilde{C} \end{pmatrix} \begin{pmatrix} d_0 \\ s_0 \\ b_0 \end{pmatrix}_R + \text{h.c.},$$

$$|\tilde{A}| = |\tilde{A}'|, \quad |\tilde{B}| = |\tilde{B}'|,$$

where u_0, c_0 , etc., denote the quark fields in the particular reference frame, in which the mass matrices have the form (2.1) (they may be called the eigenstates of the weak interaction). The parameters $A, A', \dots$ are complex. We shall assume $|A| \ll |B| \ll |C|$ and $|\tilde{A}| \ll |\tilde{B}| \ll |\tilde{C}|$. The ansatz (2.1) realizes our idea that the masses of the heavy quarks are introduced initially (they are of order $|C|$ and $|\tilde{C}|$ respectively), and the c, s quark masses and subsequently the u, d quark masses are generated by the weak interaction mixing. The mass generation proceeds like a cascade. First m_c and m_s are generated *via* the mixing described by the B parameters; subsequently m_u and m_d are generated *via* the mixing described by the A parameters.

The matrices (2.1) will be analyzed in more detail below. First we study the question under which circumstances the matrices (2.1) are obtained naturally. Due to the inequalities $|A| \ll |B| \ll |C|$ one is invited to speculate that B and A are generated as a higher-order effect, i.e., $|B| \sim (g^2/4\pi)|C|$, $|A| \sim (g^2/4\pi)|B|$ where g is the $SU(2)^w$ gauge coupling which is related to α by

$$\frac{g^2}{4\pi} = \frac{\alpha}{\sin^2 \theta_w} = \alpha_w.$$

Note that $\alpha_w \approx 27^{-1} \approx 0.04$ for $\sin^2 \theta_w \approx 0.2$.

We can express the parameters A, B, C by the quark masses. For $m_u \ll m_c \ll m_t$ and $m_d \ll m_s \ll m_b$ one has

$$\begin{aligned} |A| &= |A'| \approx \sqrt{m_u m_c}, & |\tilde{A}| &= |\tilde{A}'| \approx \sqrt{m_d m_s}, \\ |B| &= |B'| \approx \sqrt{m_t m_c}, & |\tilde{B}| &= |\tilde{B}'| \approx \sqrt{m_b m_s}, \\ |C| &\approx m_t, & |\tilde{C}| &\approx m_b. \end{aligned} \quad (2.2)$$

Since the t quark mass is unknown, we cannot calculate A, B and C . However in case of the quarks of charge $-\frac{1}{3}$ we can do so. We normalize the d and s quark

masses by setting $m_b = 1$. In this case the matrix (2.1) takes the form

$$\begin{bmatrix} 0 & \tilde{A} & 0 \\ \tilde{A}' & 0 & \tilde{B} \\ 0 & \tilde{B}' & \tilde{C} \end{bmatrix} \approx \begin{bmatrix} 0 & 0.007 & 0 \\ 0.007 & 0 & 0.18 \\ 0 & 0.18 & 1 \end{bmatrix} \approx \begin{bmatrix} 0 & 5.3\alpha_w^2 & 0 \\ 5.3\alpha_w^2 & 0 & 4.9\alpha_w \\ 0 & 4.9\alpha_w & 1 \end{bmatrix} \quad (2.3)$$

(we have left out the phase parameters). The coefficients multiplying α_w or α_w^2 are of the order of five, i.e., they are not very large compared to one, and a perturbation expression of the A and B coefficients starting with α_w^2 or α_w respectively seems not implausible.

The same pattern is apparent for the lepton masses. The (e, μ, τ) mass matrix analogous to eq. (2.3) is (all masses are normalized in units of m_τ):

$$\begin{bmatrix} 0 & 0.004 & 0 \\ 0.004 & 0 & 0.24 \\ 0 & 0.24 & 1 \end{bmatrix} \approx \begin{bmatrix} 0 & 2.9\alpha_w^2 & 0 \\ 2.9\alpha_w^2 & 0 & 6.5\alpha_w \\ 0 & 6.5\alpha_w & 1 \end{bmatrix}.$$

Under which circumstance are the matrices of the type (2.1) obtained in a natural way? In order to answer this question we study the analogous situation in the four-flavor case. Here the corresponding matrices for the (u, c) and (d, s) system are [4,5]

$$(u_0, c_0)_L \begin{pmatrix} 0 & A \\ A' & B \end{pmatrix} \begin{pmatrix} u_0 \\ c_0 \end{pmatrix}_R + \text{h.c.}, \quad (d_0, s_0)_L \begin{pmatrix} 0 & \tilde{A} \\ \tilde{A}' & \tilde{B} \end{pmatrix} \begin{pmatrix} d_0 \\ s_0 \end{pmatrix}_R + \text{h.c.},$$

$$|A| = |A'|, \quad |\tilde{A}| = |\tilde{A}'|. \quad (2.5)$$

First we notice that by a suitable unitary transformation in the space of the two right-handed or left-handed fermions $(u, c)_{R(L)}$ any 2×2 mass matrix can be written in the form

$$(u_0, c_0)_L \begin{pmatrix} 0 & A_1 \\ A_2 & B \end{pmatrix} \begin{pmatrix} u_0 \\ c_0 \end{pmatrix}_R + \text{h.c.}, \quad (2.6)$$

i.e., there are no $\bar{u}_{0L}u_{0R}$ terms. The matrix (2.5) is special in the sense that here $|A_1| = |A_2| = |A|$; the absolute value of the $\bar{u}_{0L}c_{0R}$ coupling is equal to the absolute value of the $\bar{u}_{0R}c_{0L}$ coupling. The mass matrix has a left-right symmetry. One way to arrive at matrices of the type (2.5) is to assume that the gauge group of flavor dynamics involves the group $SU(2)_L \times SU(2)_R \times U(1)$, where $SU(2)_L$ acts on the left-handed fermions, and $SU(2)_R$ acts on the right-handed fermions. A gauge group $SU(2)_L \times SU(2)_R \times U(1)$ is consistent with the present phenomenological constraints [18] provided the minimal gauge group $SU(2)_L \times U(1)$ is a subgroup which is less strongly broken than $SU(2)_L \times SU(2)_R \times U(1)$. The scalar fields which can couple to the quarks and are responsible for the generation of the quark masses must trans-

form under $SU(2)_L \times SU(2)_R$ like $(\frac{1}{2}, \frac{1}{2})$. Suppose there exist two $(\frac{1}{2}, \frac{1}{2})$ representations, which we call U and V and which we denote by two 2×2 matrices:

$$U = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix}, \quad V = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix}.$$

The most general scalar-quark interaction is of the form

$$\begin{aligned} & g_1 (\overline{u_0}, \overline{d_0})_L U \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_R + g_2 (\overline{u_0}, \overline{d_0})_L U \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_R + g_3 (\overline{c_0}, \overline{s_0})_L U \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_R \\ & + g_4 (\overline{c_0}, \overline{s_0})_L U \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_R + \text{h.c.} + (U \rightarrow V, g_1 \rightarrow g'_1, \text{etc.}) \\ & + g_5 (\overline{u_0}, \overline{d_0})_L \tau_2 U^* \tau_2 \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_R + \dots + (U^* \rightarrow V^*, g_5 \rightarrow g'_5, \text{etc.}) \end{aligned} \quad (2.7)$$

We require the scalar-quark interaction to be invariant under the following $U(1) \times U(1)$ symmetry:

$$\begin{aligned} \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_L & \rightarrow e^{i\alpha} \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_L, & \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_R & \rightarrow e^{-i\alpha} \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_R, \\ \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_L & \rightarrow e^{i\beta} \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_L, & \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_R & \rightarrow e^{-i\beta} \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_R, \\ U & \rightarrow e^{2i\alpha} U, & V & \rightarrow e^{i(\alpha+\beta)} V. \end{aligned} \quad (2.8)$$

The interaction which is invariant under the $U(1) \times U(1)$ symmetry is

$$g (\overline{c_0}, \overline{s_0})_L U \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_R + h (\overline{c_0}, \overline{s_0})_L V \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_R + h' (\overline{u_0}, \overline{d_0})_L V \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_R + \text{h.c.} \quad (2.9)$$

Under a parity transformation P the fields transform as follows:

$$\begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_{L,R} \rightarrow \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_{R,L}, \quad \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_{L,R} \rightarrow \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_{R,L}, \quad U \rightarrow U^+, \quad V \rightarrow V^+. \quad (2.10)$$

We suppose that the interaction (2.9) is invariant under P . Thus we have $g = g^*$, $g' = g'^*$, $h' = h^*$. The neutral components of the scalar fields (U_{11} , U_{22} ; V_{11} , V_{22}) acquire, in general, complex vacuum expectation values. The resulting mass matrices are of the type (2.5).

In the case of six quarks we introduce the following $U(1) \times U(1)$ symmetry:

$$\begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_L \rightarrow e^{i\alpha} \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_L, \quad \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_R \rightarrow e^{-i\alpha} \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_R,$$

$$\begin{aligned}
\begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_L &\rightarrow e^{i\beta} \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_L, & \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_R &\rightarrow e^{-i\beta} \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_R, \\
\begin{pmatrix} t_0 \\ b_0 \end{pmatrix}_L &\rightarrow e^{i(\alpha+\beta)/2} \begin{pmatrix} t_0 \\ b_0 \end{pmatrix}_L, & \begin{pmatrix} t_0 \\ b_0 \end{pmatrix}_R &\rightarrow e^{-i(\alpha+\beta)/2} \begin{pmatrix} t_0 \\ b_0 \end{pmatrix}_R, \\
U &\rightarrow e^{i(\alpha+\beta)} U, & V &\rightarrow e^{i(\alpha+3\beta)/2} V.
\end{aligned} \tag{2.11}$$

The scalar-quark interaction which respects this symmetry is

$$\begin{aligned}
&f(\overline{u_0}, \overline{d_0})_L U \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_R + f'(\overline{c_0}, \overline{s_0})_L U \begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_R + g(\overline{c_0}, \overline{s_0})_L V \begin{pmatrix} t_0 \\ b_0 \end{pmatrix}_R \\
&+ g'(\overline{t_0}, \overline{b_0})_L V \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_R + h(\overline{t_0}, \overline{b_0})_L U \begin{pmatrix} t_0 \\ b_0 \end{pmatrix}_R + \text{h.c.}
\end{aligned} \tag{2.12}$$

Furthermore we request left-right symmetry which implies $f^* = f'$, $g^* = g'$, $h^* = h$. We parametrize the vacuum expectation values as follows:

$$\langle U \rangle_0 = \begin{pmatrix} U_1 & 0 \\ 0 & U_2 \end{pmatrix}, \quad \langle V \rangle_0 = \begin{pmatrix} V_1 & 0 \\ 0 & V_2 \end{pmatrix}. \tag{2.13}$$

The mass terms which follow from (2.12) are

$$\begin{aligned}
&(\overline{u_0}, \overline{c_0}, \overline{t_0}) \begin{bmatrix} 0 & fU_1 & 0 \\ f^*U_1 & 0 & gV_1 \\ 0 & g^*V_1 & hU_1 \end{bmatrix} \begin{bmatrix} u_0 \\ c_0 \\ t_0 \end{bmatrix}_R + \text{h.c.}, \\
&(\overline{d_0}, \overline{s_0}, \overline{b_0}) \begin{bmatrix} 0 & fU_2 & 0 \\ f^*U_2 & 0 & gV_2 \\ 0 & g^*V_2 & hU_2 \end{bmatrix} \begin{bmatrix} d_0 \\ s_0 \\ b_0 \end{bmatrix}_R + \text{h.c.}
\end{aligned} \tag{2.14}$$

These mass terms are precisely of the type (2.1). However there is an additional constraint. According to eq. (2.14) we have $fU_1/fU_2 = hU_1/hU_2 = A/\tilde{A} = C/\tilde{C}$. This constraint arises since we have chosen the simplest possible symmetry breaking scheme in order to construct the mass matrices of the type (2.1), which involves the two $(\frac{1}{2}, \frac{1}{2})$ representations of the scalar fields. If we add more $(\frac{1}{2}, \frac{1}{2})$ scalars, the constraint mentioned above is not valid anymore. In ref. [9] we have used essentially the same type of symmetry breaking as discussed here, i.e., there the constraint $A/\tilde{A} = C/\tilde{C}$ is valid. We shall see that this constraint allows to calculate m_t in terms of the other quark masses [10,12] (see also ref. [19]).

3. Calculation of the Cabibbo angle

Before we discuss the six-quark case, we shall return to the four-quark case. We diagonalize the matrices given in eq. (2.5). We can rewrite them as follows:

$$(\overline{u_0}, c_0) \begin{pmatrix} 0 & a^{i\alpha} \\ ae^{i\alpha'} & be^{i\beta} \end{pmatrix} \begin{pmatrix} u_0 \\ c_0 \end{pmatrix}_R + \text{h.c.} \quad (3.1)$$

(analogously for the (d, s) system). First we redefine the quark fields:

$$u'_{0L} = e^{-i\alpha} u_{0L}, \quad c'_{0L} = e^{-i\beta} c_{0L}, \quad u'_{0R} = e^{i(\alpha' - \beta)} u_{0R}, \quad c'_{0R} = c_{0R}. \quad (3.2)$$

With these definitions the phases in the 2×2 matrix given in (3.1) are absorbed, and the mass term takes the form:

$$(\overline{u'_0}, c'_0) \begin{pmatrix} 0 & a \\ a & b \end{pmatrix} \begin{pmatrix} u'_0 \\ c'_0 \end{pmatrix}_R + \text{h.c.} \quad (3.3)$$

Subsequently we diagonalize the symmetric matrix by a common rotation among the fields $(u'_0, c'_0)_L$ and $(u'_0, c'_0)_R$, which is described by an angle θ :

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} 0 & a \\ a & b \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} -m_u & \\ & m_c \end{pmatrix}. \quad (3.4)$$

Note that we arrive at a diagonal quark mass matrix, in which the sign of m_u is negative. This simply reflects the fact that the determinant of the mass matrix given in eq. (3.3) is negative, i.e., one of the mass eigenvalues is negative. This poses no problem, since the mass of a Fermi-Dirac field ψ need not be positive. We can always change the sign of the mass term by making the transformation $\psi_R \rightarrow -\psi_R$, $\psi_L \rightarrow \psi_L$.

The angle θ can be expressed in terms of the mass eigenvalues as follows:

$$\theta = \arctan \sqrt{\frac{m_u}{m_c}},$$

$$\left(\sin \theta = \sqrt{\frac{m_u}{m_c + m_u}}, \quad \cos \theta = \sqrt{\frac{m_c}{m_c + m_u}} \right). \quad (3.5)$$

The same analysis can be carried out for the (d, s) system. In this case the rotation angle $\tilde{\theta}$ is given by

$$\arctan \tilde{\theta} = \sqrt{\frac{m_d}{m_s}}. \quad (3.6)$$

Finally we consider the weak currents. In the original basis denoted by q_0 ($q = u,$

d; c, s) the weak doublets have the form

$$\begin{pmatrix} u_0 \\ d_0 \end{pmatrix}_L, \quad \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_L. \quad (3.7)$$

In terms of the transformed fields they become:

$$\begin{aligned} \begin{pmatrix} u_0 \\ d_0 \end{pmatrix} &= \begin{pmatrix} e^{i\alpha} u'_0 \\ e^{i\tilde{\alpha}} d'_0 \end{pmatrix} = \begin{pmatrix} e^{i\alpha} (u \cos \theta + c \sin \theta) \\ e^{i\tilde{\alpha}} (u \cos \tilde{\theta} + s \sin \tilde{\theta}) \end{pmatrix}, \\ \begin{pmatrix} c_0 \\ s_0 \end{pmatrix}_L &= \begin{pmatrix} c'_0 \\ s'_0 \end{pmatrix}_L = \begin{pmatrix} -u \sin \theta + c \cos \theta \\ -d \sin \tilde{\theta} + s \cos \tilde{\theta} \end{pmatrix}_L. \end{aligned} \quad (3.8)$$

Therefore the weak coupling matrix for the mass eigenstates can be written as

$$(\overline{u}, c)_L \begin{pmatrix} e^{i(\tilde{\alpha}-\alpha)} \cos \theta \cos \tilde{\theta} + \sin \theta \sin \tilde{\theta} & e^{i(\tilde{\alpha}-\alpha)} \cos \theta \sin \tilde{\theta} - \sin \theta \cos \tilde{\theta} \\ e^{i(\tilde{\alpha}-\alpha)} \sin \theta \cos \tilde{\theta} - \cos \theta \sin \tilde{\theta} & e^{i(\tilde{\alpha}-\alpha)} \sin \theta \sin \tilde{\theta} + \cos \theta \cos \tilde{\theta} \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix}_L. \quad (3.9)$$

The weak coupling matrix can be rewritten by redefining the relative phases of the quark fields as follows:

$$(\overline{u}, c)_L \begin{pmatrix} \cos \theta_C & \sin \theta_C \\ -\sin \theta_C & \cos \theta_C \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix}_L \quad (3.10)$$

where θ_C is the Cabibbo angle. The latter is given by $\theta, \tilde{\theta}$ and the phase $\tilde{\alpha} - \alpha$:

$$\begin{aligned} \sin \theta_C &= |e^{i(\tilde{\alpha}-\alpha)} \cos \theta \sin \tilde{\theta} - \sin \theta \cos \tilde{\theta}| \\ &= \left| e^{i(\tilde{\alpha}-\alpha)} \sqrt{\frac{m_c}{m_c + m_u}} \sqrt{\frac{m_d}{m_s + m_d}} - \sqrt{\frac{m_u}{m_c + m_u}} \sqrt{\frac{m_s}{m_s + m_d}} \right|. \end{aligned} \quad (3.11)$$

Depending on the phase $(\tilde{\alpha} - \alpha)$, which cannot be calculated in terms of the quark masses, the Cabibbo angle varies between $\tilde{\theta} - \theta$ and $\tilde{\theta} + \theta$. Using the quark mass values denoted in sect. 1, one finds:

$$\begin{aligned} \theta &= \arctan \sqrt{\frac{m_u}{m_c}} \approx 3.4^\circ, \quad \tilde{\theta} = \arctan \sqrt{\frac{m_d}{m_s}} \approx 12.6^\circ, \\ 9.2^\circ &< \theta_C < 16^\circ. \end{aligned} \quad (3.12)$$

The observed value of the Cabibbo angle is well within the bound given above. Unfortunately there is no way within the approach discussed here to improve the situation, i.e., by calculating the phase parameter $\tilde{\alpha} - \alpha$. Note however that the phase ambiguity drops out in the limit $\theta \rightarrow 0$ (i.e., $m_u \rightarrow 0$). In this case the Cabibbo angle is given by $\theta_C = \arctan \sqrt{m_d/m_s}$, which agrees well with the experimental value. For this reason one may take the attitude that the weak interaction mixing proceeds

only in the (d, s) system, whereas the mass generation of the u quark has a different origin (e.g., the u quark mass may arise as a radiative correction which does not involve a mixing in the (u, c) system, see for example ref. [5]).

4. Six-quark mixing

Now we proceed to calculate the weak interaction mixing in the case of three quark doublets. We rewrite the mass terms given in eq. (2.1) as follows:

$$\overline{(u_0, c_0, t_0)}_L \begin{bmatrix} 0 & ae^{i\alpha} & 0 \\ ae^{i\alpha'} & 0 & be^{i\beta} \\ 0 & be^{i\beta'} & ce^{i\gamma} \end{bmatrix} \begin{bmatrix} u_0 \\ c_0 \\ t_0 \end{bmatrix}_R + \text{h.c.} \quad (4.1)$$

(analogously for the (d, s, b) system by replacing $a \rightarrow \tilde{a}$, etc.).

Using the quark fields $(u_0, c_0, \dots)$, the weak coupling matrix has the form:

$$\overline{(u_0, c_0, t_0)}_L \gamma_\mu \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} d_0 \\ s_0 \\ b_0 \end{bmatrix}_L. \quad (4.2)$$

Let us define new quark fields as follows:

$$\begin{aligned} u'_{0L} &= u_{0L}, & u'_{0R} &= e^{i(\alpha+\alpha'-\beta-\beta'+\gamma)} u_{0R}, \\ c'_{0L} &= e^{i(\alpha-\beta-\beta'+\gamma)} c_{0L}, & c'_{0R} &= e^{i\alpha} c_{0R}, \\ t'_{0L} &= e^{i(\alpha-\beta')} t_{0L}, & t'_{0R} &= e^{i(\alpha-\beta'+\gamma)} t_{0R}. \end{aligned} \quad (4.3)$$

In terms of these fields the mass term (4.1) takes the form

$$\overline{(u'_0, c'_0, t'_0)}_L \begin{bmatrix} 0 & a & 0 \\ a & 0 & b \\ 0 & b & c \end{bmatrix} \begin{bmatrix} u'_0 \\ c'_0 \\ t'_0 \end{bmatrix}_R + \text{h.c.} \quad (4.4)$$

Analogously we absorb the phases in the (d, s, b) system by redefining the fields d_{0L}, s_{0L} , etc. Using the new fields, the weak coupling matrix takes the form

$$(u'_0, c'_0, t'_0)_L \gamma_\mu \begin{bmatrix} 1 & 0 & 0 \\ 0 & e^{i\sigma} & 0 \\ 0 & 0 & e^{i\tau} \end{bmatrix} \begin{bmatrix} d'_0 \\ s'_0 \\ b'_0 \end{bmatrix}, \quad (4.5)$$

where

$$\sigma = (\alpha - \tilde{\alpha}) - (\beta - \tilde{\beta}) - (\beta' - \tilde{\beta}') + (\gamma - \tilde{\gamma}) ,$$

$$\tau = (\alpha - \tilde{\alpha}) - (\beta' - \tilde{\beta}') .$$

The symmetric matrix in eq. (4.4) can be diagonalized by a 3×3 rotation matrix R involving three Euler angles which denote by $\varphi_1, \varphi_2, \varphi_3$:

$$R(\varphi_1, \varphi_2, \varphi_3) \begin{bmatrix} 0 & a & 0 \\ a & 0 & b \\ 0 & b & c \end{bmatrix} R^{-1}(\varphi_1, \varphi_2, \varphi_3) = \begin{bmatrix} m_u & & \\ & -m_c & \\ & & m_t \end{bmatrix} . \quad (4.6)$$

One of the eigenvalues must be negative, since the determinant of the matrix in eq. (4.4) is negative it is equal to $-a^2 c$.

We introduce the eigenstates of the mass matrix:

$$R(\varphi_1, \varphi_2, \varphi_3) \begin{bmatrix} u'_0 \\ c'_0 \\ t'_0 \end{bmatrix} = \begin{bmatrix} u \\ c \\ t \end{bmatrix} , \quad R(\tilde{\varphi}_1, \tilde{\varphi}_2, \tilde{\varphi}_3) \begin{bmatrix} d'_0 \\ s'_0 \\ b'_0 \end{bmatrix} = \begin{bmatrix} d \\ s \\ b \end{bmatrix} . \quad (4.7)$$

In terms of the mass eigenstates the weak coupling matrix takes the following form:

$$R(\varphi_1, \varphi_2, \varphi_3) \begin{bmatrix} 1 & 0 & 0 \\ 0 & e^{i\sigma} & 0 \\ 0 & 0 & e^{i\tau} \end{bmatrix} R^{-1}(\tilde{\varphi}_1, \tilde{\varphi}_2, \tilde{\varphi}_3) . \quad (4.8)$$

The weak coupling matrix depends in particular on the two phase parameters σ and τ . Only one of them can be expressed in terms of the observable phase (see eq. (1.3)). Thus we cannot express the three weak mixing angles uniquely in terms of the quark masses; the results will depend on an unknown phase (this is analogous to the situation we encountered in the four-quark case).

First we calculate the rotation matrices R in terms of the mass eigenvalues in the limit $m_u \ll m_c \ll m_t$ and $m_d \ll m_s \ll m_b$. In this limit the R -matrices depend on only two angles, denoted here by φ_1, φ_2 , and $\tilde{\varphi}_1, \tilde{\varphi}_2$. This can be seen easily as follows. The 2×2 submatrix in eq. (4.6) involving the coefficients b and c can be diagonalised by a rotation in the (c, t) space, given by the angle $\varphi_2 = -\arctan \sqrt{m_c/m_t}$. Neglecting terms of order $(ab)/c$ we obtain the matrix:

$$\begin{bmatrix} 0 & a & 0 \\ a & -m_c & 0 \\ 0 & 0 & m_t \end{bmatrix} . \quad (4.9)$$

This matrix can be diagonalised by a rotation in the (u, c) space, given by the angle

$\varphi_1 = \arctan \sqrt{m_u/m_c}$. Thus we find:

$$R(\varphi_1, \varphi_2) = \begin{pmatrix} \cos \varphi_1 & \sin \varphi_1 & 0 \\ -\sin \varphi_1 & \cos \varphi_1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi_2 & \sin \varphi_2 \\ 0 & -\sin \varphi_2 & \cos \varphi_2 \end{pmatrix}. \quad (4.10)$$

Analogously the rotation matrix $R(\tilde{\varphi}_1, \tilde{\varphi}_2)$ is given by the angles $\tilde{\varphi}_2 = -\arctan \sqrt{m_s/m_b}$ and $\tilde{\varphi}_1 = \arctan \sqrt{m_d/m_s}$. We use the definitions

$$\begin{aligned} \mu_1 &= \sqrt{\frac{m_u}{m_c}}, & \mu_2 &= \sqrt{\frac{m_c}{m_t}}, \\ \nu_1 &= \sqrt{\frac{m_d}{m_s}}, & \nu_2 &= \sqrt{\frac{m_s}{m_b}}, \end{aligned} \quad (4.11)$$

and set $\cos \varphi_i \approx \cos \tilde{\varphi}_i \approx 1$. The weak coupling matrix given by eq. (4.8) takes the form:

$$\begin{pmatrix} 1 & -\nu_1 + \mu_1 e^{i\sigma} & \mu_1(\nu_2 e^{i\sigma} - \mu_2 e^{i\tau}) \\ -\mu_1 + \nu_1 e^{i\sigma} & \mu_1 \nu_1 + \mu_2 \nu_2 e^{i\sigma} + e^{i\sigma} & \nu_2 e^{i\sigma} - \mu_2 e^{i\tau} \\ (\mu_2 e^{i\sigma} - \nu_2 e^{i\tau})\nu_1 & \mu_2 e^{i\sigma} - \nu_2 e^{i\tau} & \mu_2 \nu_2 e^{i\sigma} + e^{i\tau} \end{pmatrix}. \quad (4.12)$$

After suitable redefinitions of the phases of the quark fields this matrix can be brought into the form given in eq. (1.3). The absolute values of the matrix elements of the matrix (4.12) are equal to the corresponding matrix elements of the matrix given in eq. (1.3). Thus we have

$$\begin{aligned} s_1 &= |\nu_1 - \mu_1 e^{i\sigma}|, \\ s_1 s_3 &= \mu_1 |\nu_2 e^{i\sigma} - \mu_2 e^{i\tau}|, \\ s_1 s_2 &= \nu_1 |\mu_2 e^{i\sigma} - \nu_2 e^{i\tau}|, \\ |s_3 + s_2 e^{i\delta}| &= |\nu_2 e^{i\sigma} - \mu_2 e^{i\tau}|, \end{aligned} \quad (4.13)$$

and we find

$$\begin{aligned} \delta &= \pi - \sigma \\ s_1 &= |\sqrt{m_d/m_s} + e^{i\delta} \sqrt{m_u/m_c}|, \\ s_2 &= \frac{\sqrt{m_d/m_s}}{s_1} r, & s_3 &= \frac{\sqrt{m_u/m_c}}{s_1} r, \\ r &= |\sqrt{m_c/m_t} + \sqrt{m_s/m_b} e^{-i(\tau+\delta)}|. \end{aligned} \quad (4.14)$$

We note that the Cabibbo angle is determined by the four quark masses m_u, m_c ;

m_d , m_s , and the phase parameter δ . The unknown phase parameter τ enters in the expressions for s_2 and s_3 .

5. The mass of the t quark

Thus far we have investigated the consequences of the mass matrices (2.1) for the mixing angles. However in the minimal scheme for the scalar fields (two $(\frac{1}{2}, \frac{1}{2})$ representations) discussed in sect. 2 there exists an additional constraint, namely

$$\left| \frac{A}{\tilde{A}} \right| = \frac{a}{\tilde{a}} = \left| \frac{C}{\tilde{C}} \right| = \frac{c}{\tilde{c}} . \quad (5.1)$$

The parameters a , $\tilde{a}$, c and $\tilde{c}$ can be calculated in terms of the quark masses. The determinant of the matrix in eq. (4.4) is equal to $-a^2 c = -m_u m_c m_u$. The trace of

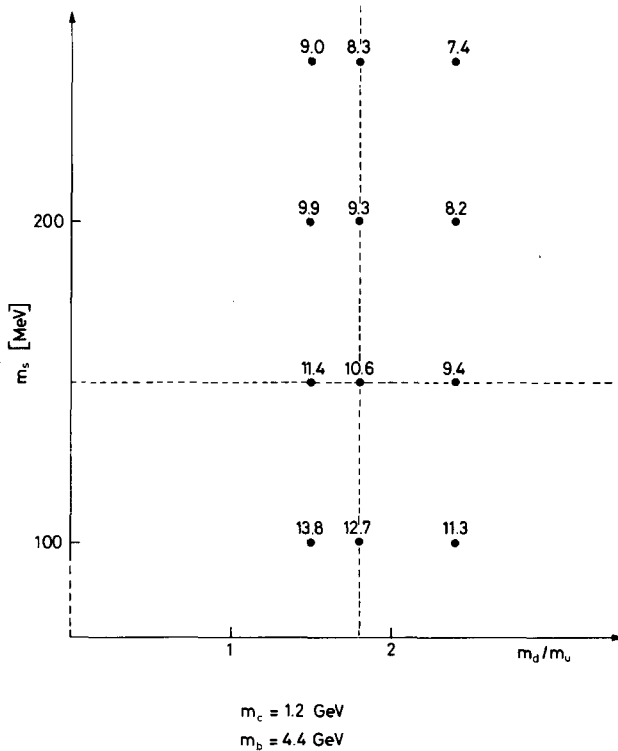


Fig. 2. The t quark mass as a function of m_d/m_u and m_s .

the same matrix is given by $c = m_t - m_c + m_u$. Thus we find

$$a = \sqrt{\frac{m_t m_c m_u}{m_t - m_c + m_u}}, \quad \tilde{a} = \sqrt{\frac{m_b m_s m_d}{m_b - m_s + m_d}}, \quad (5.2)$$

and we arrive at the relation

$$\frac{m_t m_c m_u}{m_b m_s m_d} = \left(\frac{m_t - m_c + m_u}{m_b - m_s + m_d} \right)^3.$$

This relation is obtained in the scheme discussed in ref. [9]. It was first written down in ref. [10] and ref. [12] (see above, ref. [19]). For $m_t \gg m_c \gg m_u$ and $m_b \gg m_s \gg m_d$, eq. (5.3) can be written (in a reasonable approximation) as

$$\frac{m_t}{m_b} \approx \sqrt{\frac{m_c m_u}{m_s m_d}} \approx 2.1.$$

For the quark mass values given in (1.1) one finds

$$m_t \approx 10.6 \text{ GeV}.$$

The value of the t quark mass is rather sensitive to the ratio m_d/m_u . Typical estimates of this ratio range between 1.5 and 2.5. Furthermore there exists some uncertainty about the s quark mass. For this reason we have calculated the t mass as a function of m_s and (m_d/m_u) , which is shown in fig. 2. The range of possible masses extends from about 8 GeV to 14 GeV. The most probable value is between 10 and 11 GeV. We note that the value of the t quark mass does not change much if we use in eq. (5.3) the “constituent quark masses” instead. For $m_u \approx m_d \approx 0.3$ GeV, $m_s \approx 0.5$ GeV, $m_c \approx 1.5$ GeV and $m_b \approx 4.7$ GeV one finds $m_t \approx 9.6$ GeV.

6. Weak mixing angles

Let us discuss the numerical values of the mixing angles, given by eq. (4.14). They depend in particular on the phase parameters δ and τ . However the ratio s_3/s_2 is independent of the phases. We obtain

$$\frac{s_3}{s_2} \approx \sqrt{\frac{m_u m_s}{m_c m_d}} \approx \frac{\theta_3}{\theta_2}, \quad (6.1)$$

which is 0.26 for the quark masses given in eq. (1.1). Since the angle s_3 determines the strength of the $(\bar{u}b)$ transition and s_2 the strength of the $(\bar{t}d)$ transition, we expect:

$$\frac{\text{rate}(b \rightarrow u)}{\text{rate}(t \rightarrow d)} = \frac{m_u m_s}{m_c m_d} \approx 0.07. \quad (6.2)$$

The angle θ_2 is expected to be less than $(m_c/m_t)^{1/2}$, i.e., we have $\theta_2 \lesssim 0.33$

($\lesssim 19^\circ$). Thus we find the useful constraint:

$$\theta_3 \lesssim 0.09 (\lesssim 5^\circ). \quad (6.3)$$

The parameter r can vary between its minimum and maximum value, according to eq. (6.3):

$$\left| \sqrt{\frac{m_c}{m_t}} - \sqrt{\frac{m_s}{m_b}} \right| \lesssim r \lesssim \sqrt{\frac{m_c}{m_t}} + \sqrt{\frac{m_s}{m_b}}. \quad (6.4)$$

Using the quark mass values given in eq. (1.1) and our estimate for the t quark mass, we find:

$$0.15 \lesssim r \lesssim 0.52, \quad (6.5)$$

which implies

$$9^\circ \lesssim \theta_2 \lesssim 32^\circ, \quad 2^\circ \lesssim \theta_3 \lesssim 9^\circ. \quad (6.6)$$

Together with the constraint, eq. (6.3), we find:

$$9^\circ \lesssim \theta_2 \lesssim 19^\circ, \quad 2^\circ \lesssim \theta_3 \lesssim 5^\circ. \quad (6.7)$$

It is interesting to study in particular the case $r = r_{\min}$, where both θ_2 and θ_3 reach their minimum values. In this case the parameter δ (determined by the observed magnitude of the CP violation) is given by eq. (1.4): $\delta \approx 0.12$. It is interesting that δ is of the same order on the weak mixing angles θ_i , i.e., the observed weakness of the CP violation is attributed to the fact that all weak mixing angles are small (and similar in their orders of magnitude). The CP violation in the decays of t and b flavored particles is expected to be much larger than the CP violation observed in the $K-\bar{K}$ system.

The Cabibbo angle is given by

$$s_1 \approx \left(\frac{m_d}{m_s} + \frac{m_u}{m_c} + 2 \cos \delta \sqrt{\frac{m_u}{m_c}} \sqrt{\frac{m_d}{m_s}} \right)^{1/2}. \quad (6.8)$$

Depending on the magnitude of δ , θ_1 varies between 9.5° and 16° . The observed value of θ_C (13°) is reproduced for $\delta \approx 90^\circ$. Since we have $\delta \approx 0.12$, the case $\delta \approx 90^\circ$ seems excluded. Thus the quark mass values given in eq. (1.1) combined with eq. (4.14) do not reproduce exactly the observed value of the Cabibbo angle. However we emphasize that in particular the ratio m_d/m_s is somewhat uncertain (for a discussion of this point, see e.g., ref. [4]). The ratios m_d/m_s and m_u/m_s may indeed be smaller than the ones given in eq. (1.1), and we may lower the u and d quark masses such that the ratio m_u/m_d remains unchanged (note that the ratio m_u/m_d is determined with somewhat greater accuracy than the ratio m_d/m_s). For example in the case $m_s \approx 150$ MeV, $m_u \approx 2.7$ MeV, $m_d \approx 5$ MeV we find $\theta_1 \approx 13^\circ$. We emphasize that a rescaling of the u and d quark masses such that m_u/m_d remains constant does not influence the result for m_t (see eq. (5.3)).

It is instructive to study the weak coupling matrix in the particular case in which

all the constraints discussed here are obeyed. We take $\theta_1 = \theta_c = 13^\circ$, $r = r_{\min} = 0.15$, $\theta_2 = 7^\circ$ (obtained using eq. (4.14) and $m_u = 5$ MeV), $\theta_3 = 2^\circ$ (obtained using eq. (4.14) and $m_d = 2.7$ MeV), $\delta = 0.16$ (obtained using eq. (1.4)), and find for the weak doublets

$$\begin{pmatrix} u \\ 0.97d + 0.22s + 0.008b \end{pmatrix}, \quad \begin{pmatrix} c \\ -0.22d + (0.97 - 0.0007i)s + (0.16 + 0.02i)b \end{pmatrix}, \\ \begin{pmatrix} t \\ 0.03d - (0.15 + 0.006i)s + (0.98 + 0.16i)b \end{pmatrix}. \quad (6.9)$$

We note that the violation of the Cabibbo universality in the (u, d, s) system is extremely small. One finds: $|c_1^2 + s_1^2 c_3^2 - 1| \approx 10^{-4}$.

The main weak decays of the t and b flavored particles proceed sequentially. The B particles (quark composition $(\bar{u}b)$ or $(\bar{d}b)$) decay essentially into charmed particles. One has

$$\frac{\text{rate}(b \rightarrow c)}{\text{rate}(b \rightarrow u)} \approx 400.$$

The life-time of the B particles is expected to be reduced by a factor 40, compared to the canonical life time of about 10^{-15} s, i.e., we expect $\tau(B) \approx \text{few } 10^{-14}$ s.

7. Conclusions

In this paper we have shown that the weak mixing angles can be estimated in terms of the quark masses, if the flavor dynamics is given by a $SU(2) \times SU(2) \times U(1)$ theory and if the basic interaction is left-right symmetric (parity is broken spontaneously). Furthermore within the simplest ansatz for the Yukawa interactions of the scalar fields with the fermions the mass of the t quark is given in terms of the other quark masses. We expect $m_t \approx 10 \dots 11$ GeV.

The interaction of the scalars with the fermions induces flavor-changing neutral current effects [20]. The size of those effects depend on the masses of the scalar bosons. If the latter are of the same order of magnitude as the masses of the W_L bosons, the flavor-changing neutral current effects (e.g., induced $K_1 - K_2$ mass difference) are much smaller than the observed effects or the existing limits on those effects.

The scheme of flavor dynamics discussed here is naturally embedded together with the QCD of the strong interactions in the grand unified theory based on the group $SO(10)$ [21] (for a recent discussion see ref. [22]). Thus far, we have not discussed at which energy the left-right symmetric flavor theory based on $SU(2)_L \times SU(2)_R \times U(1)$ becomes relevant, i.e. at which energy the effects of the gauge bosons belonging to the group $SU(2)_R$ set in. Various phenomenological constraints require $M(W)_R \gg M(W)_L$. It may be that the masses of the bosons belonging to

$SU(2)_R$ are superheavy, i.e., that they are generated by the primary symmetry breaking of the grand unified theory (e.g., the theory based on $SO(10)$). If this is the case, the symmetry breaking in the $SO(10)$ theory proceeds as follows:

$$\begin{array}{c}
 SO(10) \\
 \downarrow \text{superheavy symmetry breaking} \\
 \quad \text{(including breaking of parity)} \\
 SU(3)^c \times SU(2) \times U(1) \\
 \downarrow \text{ordinary symmetry breaking} \\
 SU(3)^c \times U(1)^e .
 \end{array}$$

The consequences of such a theory would be very similar to the ones obtained in the $SU(5)$ approach [23], except one would find the additional results discussed in this paper.

After completion of this paper we received a preprint by Shrock, Treiman and Wang [25], in which new, improved bounds on the mixing angles were derived, based on phenomenological constraints (Cabibbo universality, observed magnitude of CP violation, $K_1 - K_2$ mass difference). They find in the case of $m_t \sim 10$ GeV:

$$0.1 \leq s_2 \leq 0.4, \quad (\text{for } \text{sgn}(\cos \delta) = +1),$$

$$0.4 \leq s_2 \leq 0.7, \quad (\text{for } \text{sgn}(\cos \delta) = -1).$$

The first bound is essentially identical to the one used in sect. 6. However, the second case allows a larger value of θ_2 and would imply (according to eq. (6.1)):

$$0.10 \leq s_3 \leq 0.18, \quad (6^\circ \leq \theta_3 \leq 10^\circ).$$

I would like to thank J. Ellis, H. Georgi, H. Leutwyler, P. Minkowski, R. Peccei and K. Tanaka for useful discussions. After completion of this work I received a preprint of Georgi and Nanopoulos [24], in which some of the results discussed here were derived within the grand unified theory based on $SO(10)$. Furthermore I would like to thank Dr. Shrock for informing me about the results of ref. [25].

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